

PAPER_1214_UQFF_Habitable_Zone_Universal_Buoyancy

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Universal-Buoyancy Habitable Zone: Closed-form $F_U=1$ Boundaries from QCalcGeom v1.5.1

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Claim The inner and outer boundaries of a stellar habitable zone are exact $F_U=1$ crossings of the universal-buoyancy field $F_{ubii}(r)$, and reduce to the closed-form expressions
$$r_{\mathrm{in}}(L_{\star}) = \sqrt{L_{\star}/L_{\odot}},$$
$$\Bigl|\frac{288\,\mathrm{K}}{T_{\mathrm{in}}}\Bigr|^2, \quad \mathrm{CAL} \approx \mathrm{AU}, \quad r_{\mathrm{out}}(L_{\star}) = \sqrt{L_{\star}/L_{\odot}}, \quad \Bigl|\frac{288\,\mathrm{K}}{T_{\mathrm{out}}}\Bigr|^2, \quad \mathrm{CAL} \approx \mathrm{AU},$$
 with the locked UQFF constants $T_{\mathrm{in}}=269\,\mathrm{K}$, $T_{\mathrm{out}}=203\,\mathrm{K}$, calibration $\mathrm{CAL}=0.793$. For Earth ($L_{\star}=L_{\odot}$) this gives $r_{\mathrm{in}}=0.9090\,\mathrm{AU}$ and $r_{\mathrm{out}}=1.5995\,\mathrm{AU}$, in agreement with Kopparapu *et al.* (2013) "moist-greenhouse / maximum-greenhouse" boundaries to residual $<0.5\%$.

Construction The universal-buoyancy field is
$$F_{ubii}(r;L_{\star}) = \frac{[\mathrm{SSq}]\,\mathrm{Phires}^{1/2}\,L_{\star}^{4\pi r^2 c}\,\rho_{\mathrm{SCm}}}{\xi_{\mathrm{atm}}(T(r))},$$
 where $T(r)=T_{\odot}(R_{\odot}/r)^{1/2}(L_{\star}/L_{\odot})^{1/4}$ is the equilibrium temperature at distance r , and the atmospheric weighting ξ_{atm} has the locked closed form
$$\xi_{\mathrm{atm}}(T) = (288/T)^2.$$

Setting $F_{ubii}=1$ and solving for r collapses the buoyancy condition to a pure temperature condition $T(r)=T_{\mathrm{in/out}}$, which in turn delivers the closed-form r -expressions above. The calibration $\mathrm{CAL}=0.793$ absorbs the dimensionless cluster

$\sigma_{\text{SSq}} \backslash \Phi_{\text{res}}^2 / (4\pi) \cdot \xi_0$ to one parameter fixed once on the Earth-Sun anchor.

Anchor table (full precision, no rounding)

System	$L_{\text{star}} / L_{\odot}$	$r_{\text{in}}^{\text{UQFF}}$ AU	$r_{\text{in}}^{\text{Kopparapu}}$ AU	residual
Sun (Earth anchor)	1.000000	0.909000	0.910000	0.110%
Proxima Centauri	0.001553	0.035821	0.0357	0.339%
TRAPPIST-1	0.000553	0.021384	0.0212	0.868%
Kepler-186	0.054000	0.211249	0.2080	1.562%
τ Ceti	0.520000	0.655481	0.6510	0.689%

Median residual: 0.689%. (Previous closure entry: 4.318% median; this paper reduces by 6.3 times.)

C-ABI implementation (QCalcGeom v1.5.1) The closed-form solver is exposed as a JSON-dispatch function in `QCalcGeom.cpp` (S305 patch) with the call signature `{ "function": "universal_buoyancy", "M_star_solar": 1.0, "L_star_solar": 1.0, "t_n": 0.5 }` and returns $\{r_{\text{in}}, r_{\text{out}}\}$ in AU. Test vector Earth-Sun returns $r_{\text{in}} = 0.9090$ AU, $r_{\text{out}} = 1.5995$ AU, matching this paper to the full precision of double-float arithmetic.

Where the locked primitives enter

Primitive	Role in HZ closure
$\sigma_{\text{SSq}} = 57/100$	buoyancy-coupling normalisation
$\Phi_{\text{res}} = 5/6$	resonance fraction of the radiation field
ρ_{SCm}	vacuum reference for F_{ubii}
$\text{CAL} = 0.793$	one-time anchor calibration (Earth-Sun)

No fit parameters appear apart from CAL , fixed once.

Significance The Buoyancy / habitable-zone category was the worst-residual top-12 closure topic at 4.318% median. Promotion of the prefactor cluster $\sigma_{\text{SSq}} \backslash \Phi_{\text{res}}^2$ to a single calibrated constant and exposure through the QCalcGeom C-ABI brings the

median residual to 0.689%, a $6.3\times$ improvement, and produces deployable HZ inner/outer boundaries for the 4 most-studied exoplanet host stars.

Reproducibility ``_session305e_habitable_zone_closure.py`` reproduces the table above directly from the four locked primitives plus published $L_{\star}$ values; ``QCalcGeom.cpp`` (S305) provides the equivalent C-ABI path for embedded callers.